*Candidate's examination number……..*

**PRESIDENT'S OFFICE**

**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**

**NAMANGA SECONDARY SCHOOL**

**FORM TWO PRE-MOCK EXAMINATION**

**BIOLOGY-AUGUST-2019**

**CODE: 0033** **Time 2:30Hours**

## INSTRUCTIONS

1. This paper consists of section A, B, and **C** with a total of **eleven (11)** questions.
2. Answer ALL questions in section A and B and **one (1)** question from section C.
3. All writing must be in blue or black ink except diagrams which must be drawn by pencil and label by ink.
4. All means of communication appliances and calculators are not allowed in examination room.
5. Write your examination Number at the top right corner of every page.

| FOR EXAMINER'S USE ONLY | | |
|---|---|---|
| **QUESTION NUMBER** | **SCORE** | **INITIALS FOR EXAMINER** |
| 1. | | |
| 2. | | |
| 3. | | |
| 4. | | |
| 5. | | |
| 6. | | |
| 7. | | |
| 8. | | |
| 9. | | |
| 10. | | |
| 11. | | |
| TOTAL | | |

**SECTION A (35 Marks)**

**Answer all questions in this section.**

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter beside item number in the box provided.

   i. An experiment usually tests:-
      A. Application.
      B. Evaluation
      C. Hypothesis
      D. Problem identification

   ii. Which one is out of place about importance of First Aid?
      A. Brings hope and encouragement to the patient
      B. Improves the victim's life span
      C. Reduces fear of death
      D. Prevents shock

   iii. Which of the following are communicable diseases?
      A. Cancer, Gonorrhea and Syphilis.
      B. Cholera, Gonorrhea and Syphilis
      C. Cholera. Kwashiorkor and Marasmus
      D. Cholera, Marasmus and Rickets

   iv. In the human digestive system, a stomach is known as:-
      A. A tissue
      B. A cell
      C. An organ
      D. An organism

   v. An organism with a rigid protective wall made up of chitin material is a representative of which kingdom?
      A. Fungi
      B. Monera
      C. Plantae
      D. Protoctista

vi. Jasmine cannot see objects in a dim light. What condition is she probably suffering from?

A. Night blindness
B. Kwashiorkor
C. Beriberi
D. No carrot in her diet

vii. The natural environment includes:-

A. predators and preys
B. plants and animals
C. consumers and producers
D. living and non-living things

viii. Diffusion is:-

A. Going from adjacent area to its gradient area
B. Going from an area of low to high concentration
C. Going from water to the solute
D. Going from an area of high to low concentration

ix. Where does gaseous exchange take place in fish?

A. Gills
B. Lungs
C. Trachea
D. Skin

x. The structure of the leaf that allows oxygen and carbon dioxide to diffuse into and out of the leaf during gaseous exchange is called:-

A. Cuticle
B. Epidermis
C. Stomata
D. Mesophyll.

2. For each of the items (i) – (x), write **TRUE** if a statement is correct or **FALSE** if a statement is not correct in a space provided.
   i. Dark blue color determines the presence of starch in a solution when iodine solution is added ……..
   ii. A person with a blood group AB is called Universal donor ……..
   iii. An incinerator is used to dispose hazardous wastes ……..
   iv. Examples of animal organs are the skin, eyes, roots and fruits ……
   v. Soil and water represent abiotic factor ……..
   vi. People living with HIV/AIDS should be stigmatized ……
   vii. Biology is the study of non living things …….
   viii. The largest group in the classification is phylum …….
   ix. Transpiration is the not a necessary process to plants ……
   x. An onion stores food in leaves while carrots store food in roots ……
3. Match the items in **List A** with responses in **list B** by writing the letter of the correct response beside the item number on the answer sheet provided.

| **LIST A** | **LIST B** |
|---|---|
| i. Donate blood to all groups | A. Thallus |
| ii. The type of immunity which a person born with it | B. Sorus |
| iii. An example of protozoan diseases | C. Food chain |
| iv. Bacteria which cause typhoid fever | D. Food web |
| v. Plant cell which do not have chloroplast | E. Salmonella typhii |
| vi. A group of cell joined together to perform particular function | F. Pesturella pestis |
| vii. Amoeba, paramecium and Euglena | G. Carbon dioxide |
| viii. Flat body of liverworsts | H. Oxygen |
| ix. A network of food chain | I. Tissue |
| x. Gases produced during photosynthesis | J. System |
| | K. Leaf cell |
| | L. Root hair cell |
| | M. Fungi |
| | N. Protoctista |
| | O. Natural immunity |
| | P. Artificial immunity |
| | Q. Blood group AB |
| | R. Blood group O |
| | S. Amoebic dysentery |

| **LIST A** | I | ii | iii | iv | v |
|---|---|---|---|---|---|

| **LIST B** | | | | | |
|---|---|---|---|---|---|

4. Complete each of the following statements by writing the correct answer in the space provided in each item number of question.
   i. The part of the cell which controls movement of materials in and out of the cell during life process of an organism……..
   ii. A cell in human being that has ability to move …….
   iii. The process by which a less specialized cell becomes a more specialized cell type for specific function …….
   iv. The green pigment in chloroplasts that gives leaves the green color is known as ……
   v. The system that transports oxygen and nutrients in the body of living organism is called ……

**SECTION B (50Marks)**

**Answer all questions in this section**

5. (a) Define the following terms:-
   i. Gaseous exchange.

   _____

   _____

   ii. Respiration.

   _____

   _____

   (b) Give one(1) reason for each of the following characteristics of respiratory surface.
   i. Have thin walls

   _____

   _____

   ii. They are moist.

   _____

   _____

   (c) Compare any three (3) features of aerobic respiration and anaerobic respiration.

   (i) _____

(ii) _____

(iii)_____

6. (a) Define the following terms as used in balance of nature.

i. Producers

_____

_____

ii. Consumers

_____

_____

iii. Decomposers

_____

_____

(b) Study the following food web (in figure 1) then answer the questions that follow:-

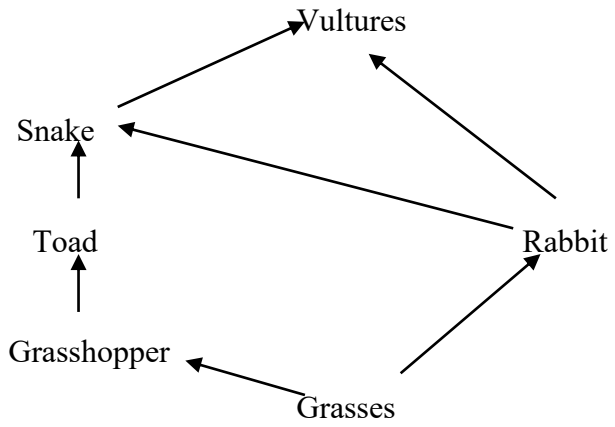



Figure 1

What would happen if:-

i. All the snakes were killed in figure 1 above?

_____

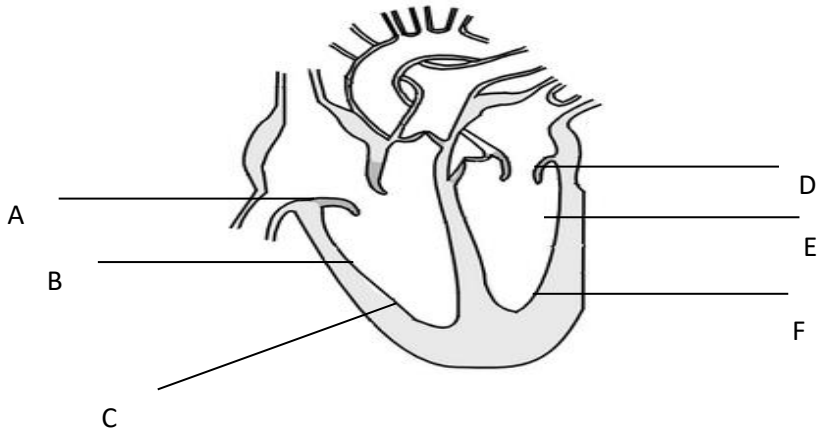
_____

ii. The number of toads multiplied?

_____

_____

7. The figure 2. Below represents a section through a mammalian heart. Study the diagram then answer the questions that follow.




(a) Name the parts labeled with A, B,C, D, E and F

A. _____

B. _____

C. _____

D. _____

E. _____

F. _____

(b) (i) State the difference between valve labeled A and D(Three points)

_____

_____

_____

_____

(ii) Why the chamber labeled F has thicker muscular walls than the chamber labeled B.

_____

_____

_____

_____

8. (a) what do you understand by the following terms:-

i. Personal hygiene.

_____

_____

_____

ii. Good manner.

_____

_____

_____

(b) State four (4) principles of:-

i. Personal hygiene

(a) _____

(b) _____

(c) _____

(d) _____

ii. Good manners

(a) _____

(b) _____

(c) _____

(d) _____

9. (a) i) what is classification?

_____

_____

_____

ii) Give two differences between natural system and artificial system of classification.

| Natural system | | Artificial system |
|---|---|---|
| (i) | | |
| (ii) | | |

(b) i) Kingdom Fungi has three main phyla. Identify them and give one example of each phylum.

ii) State at least four (4) useful economic importance of Fungi.

(a) _____

(b) _____

(c) _____

(d) _____

**SECTION C (15 MARKS)**

Answer one (1) question from this section.

10. Use the guidelines given below to explain how you will teach people in your community about eating balanced diet.
    - Meaning of the balanced diet.
    - Components of balanced diet.
    - One function of each component.
    - Deficiency of disease that may result when a person does not eat a balanced diet.
    - Two factors that contribute lack of balanced diet among children in your community.

11. Explain the causes, mode of transmission and prevention measures of typhoid one of the communicable disease in Tanzania.

**Answer**